



Code + paper

Beyond

Better-than-Full-Precision KV Caches

via Learnable Non-Uniform Quantization as Implicit Regularization



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Learn the quantizer. Freeze the model. Decode packed KV.

3.2×

KV payload compression
4-bit codes; group size 32

2.2–6.9%

Lower held-out PPL
versus full-precision Native

1.50×

Peak decode speedup
kernel-only vs FlashAttention-4

1.17×

Peak serving throughput
end-to-end vLLM vs FA4

01 Learn from language-model loss

Local KV reconstruction error is only a proxy for generation quality: perturbations propagate through attention and the remaining network.

Beyond learns representation levels R and ordered thresholds T directly from **next-token negative log-likelihood (NLL)**.

$$\min_{R,T} \mathbb{E}_X \left[- \sum_t \log p_\theta(x_{t+1} | x_{\leq t}; Q_{R,T}) \right]$$

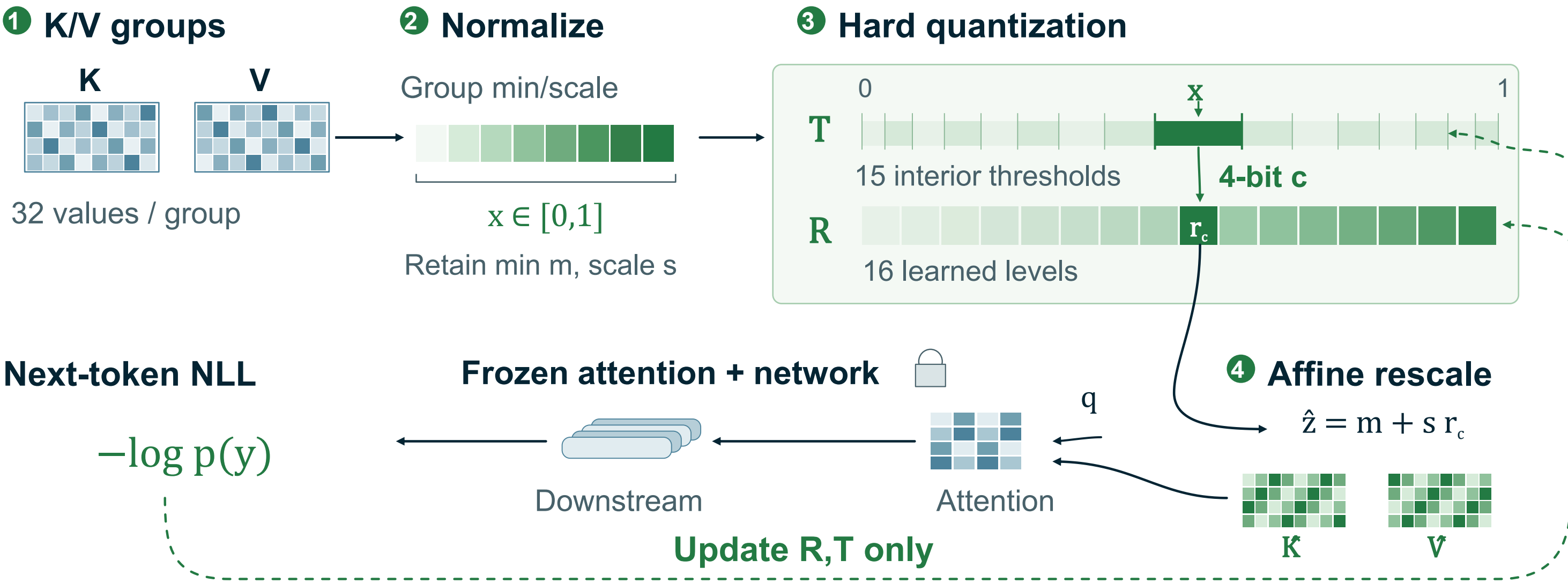
Backbone weights θ remain frozen.

$$Q(x; R, T) = r_k, \quad t_k \leq x < t_{k+1}$$

Here $x \in [0, 1]$ is group-normalized; the final bin includes $x = 1$. Training and inference use the **same hard bin assignment**. STE input gradients [1] and our boundary-localized threshold pseudo-gradients enable learning.

Hard forward pass, loss-driven updates

Quantizers vary by layer, K/V, head and group (illustrative).



One 32-value group:

128 code bits

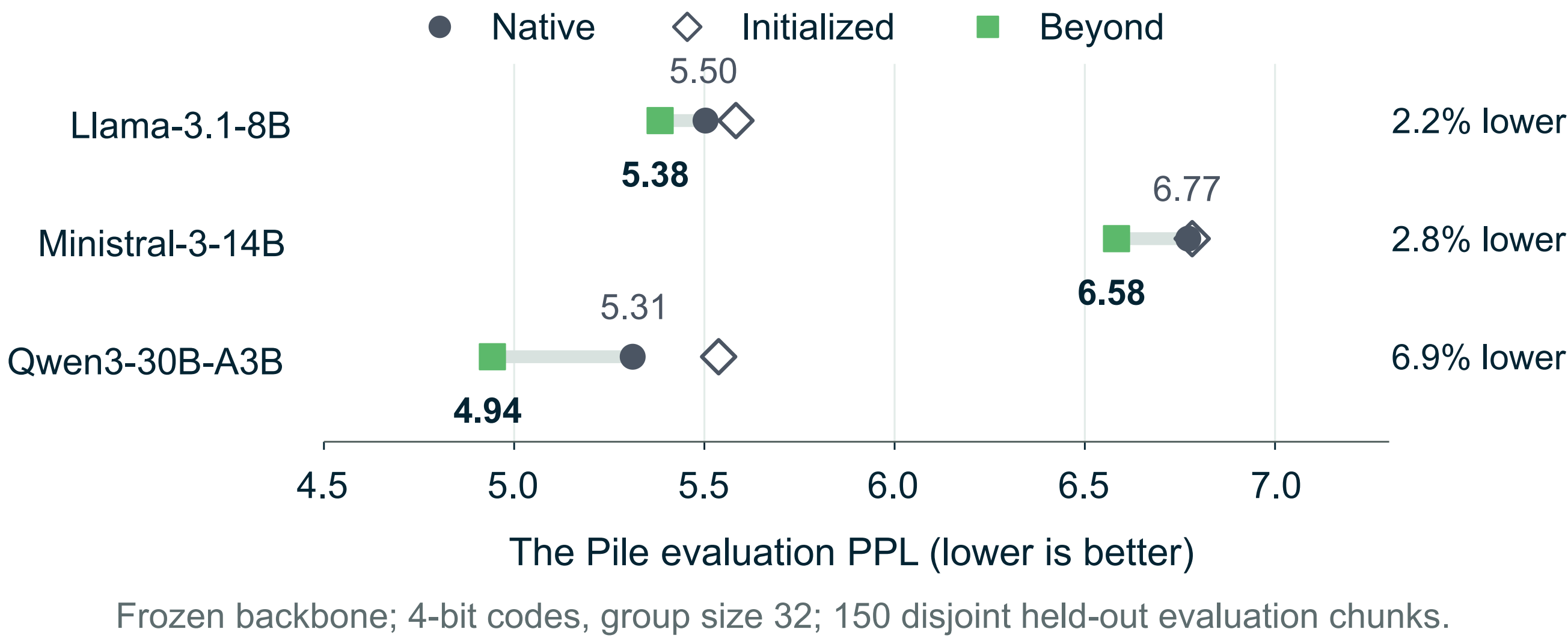
32 metadata bits

5.0 effective bits/scalar, including min/scale metadata; 3.2× compression before page padding.

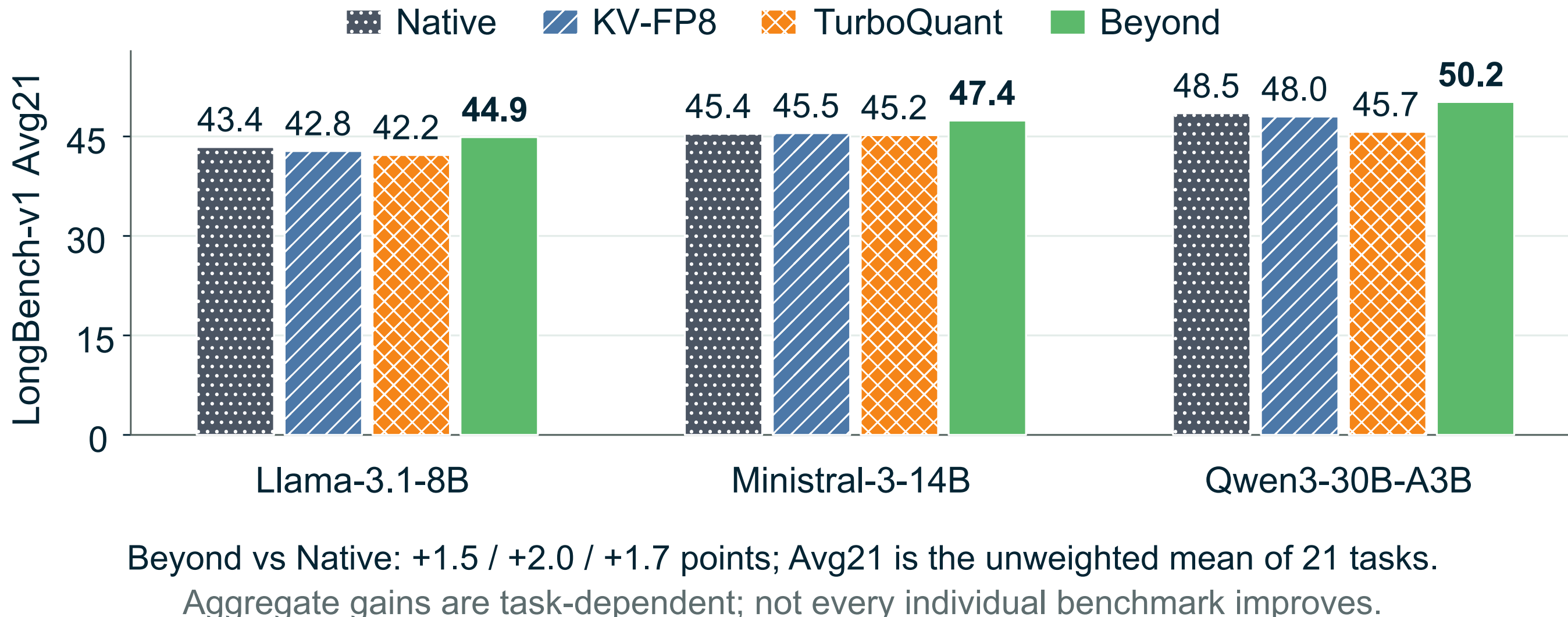
02 Quality beyond the full-precision baseline

Three instruction-tuned LLMs; 4-bit codes, G = 32; 150 held-out Pile evaluation chunks.

Quantizer-only training lowers held-out perplexity



Long-context quality improves on all three model averages

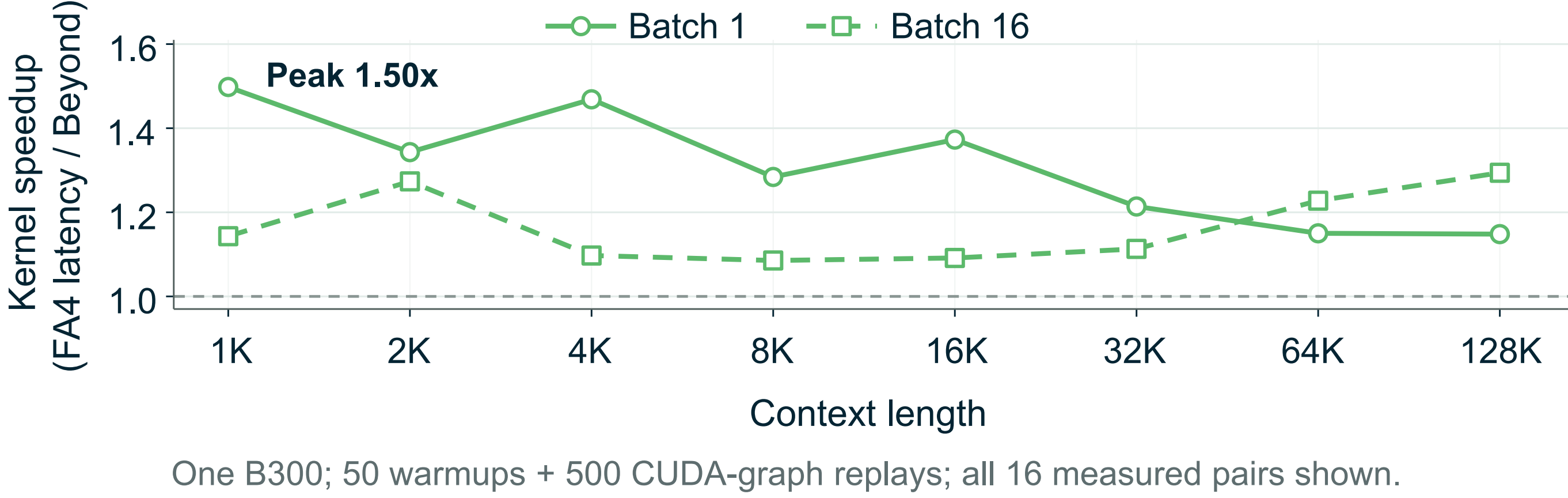


LongBench Avg21 is the unweighted mean over 21 tasks [2]. Beyond has the **best average** among the tested methods, not the best score on every task.

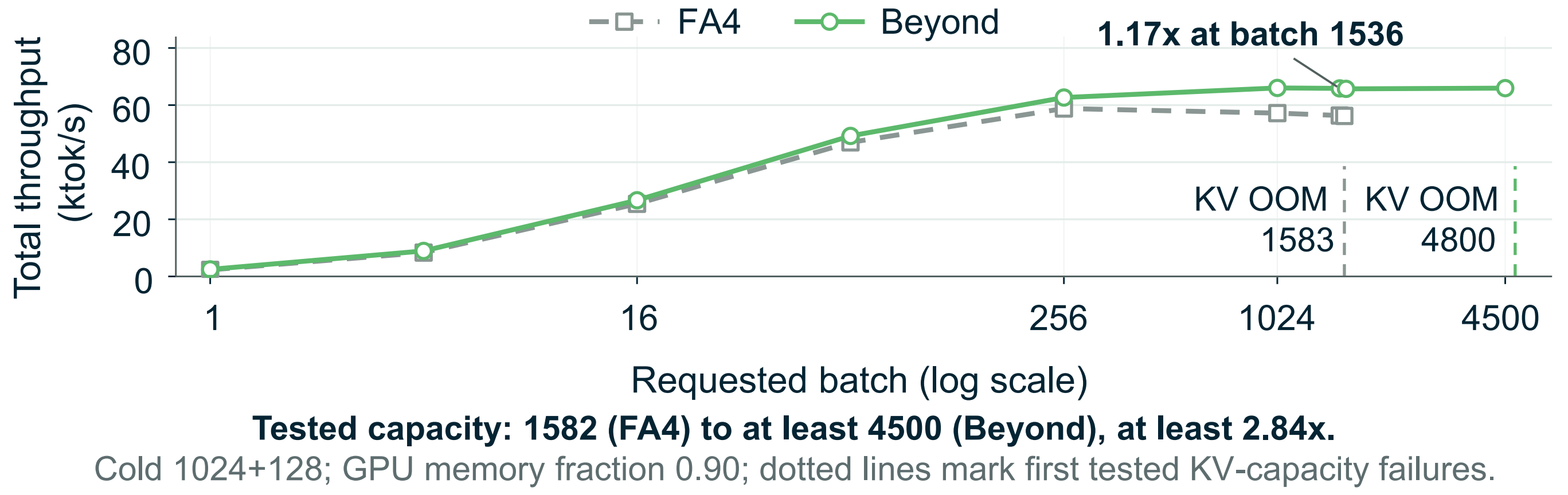
03 Packed execution, measured on NVIDIA B300

A **fused CuTe decode kernel** reads packed codes and fuses lookup, rescaling, attention scoring, online softmax and value accumulation.

Fused decode is faster than FlashAttention-4



Serving: higher throughput and larger tested batch capacity



Llama-3.1-8B-Instruct geometry; batch 1/16; 1K–128K contexts. 50 warmups + 500 CUDA-graph replays. FA4 [4] is the 1.0× reference.

Llama-3.1-8B-Instruct, BF16; official vLLM [3] benchmark: cold 1024 + 128 tokens; scheduling budget 8192; prefix caching off; GPU-memory fraction 0.9.

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[1] Bengio et al. (2013). Estimating or Propagating Gradients. arXiv:1308.3432.
[2] Bai et al. (2024). LongBench. ACL.
[3] Kwon et al. (2023). PagedAttention / vLLM. SOSP.
[4] Zadouri et al. (2026). FlashAttention-4. arXiv:2603.05451.

github.com/wsxbjnb1/Beyond-KV-Cache
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